

BCP Forces a Faithful State and an Essential Blocker Ideal

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving the commutative unital criterion for $C(K)$, it asks for a noncommutative C^* -algebra characterization:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Background

We use two previous lane-19 packets.

First,

2311.14257_bcp_countable_capture_exact_equivalence

proved that a normed space fails BCP if and only if it has countable capture. Equivalently, a normed space X has BCP if and only if there is a countable set $C \subset X$ such that for every $u \in S_X$ some $c \in C$ satisfies

$$\text{dist}(c, \mathbb{C}u) < \|c\|.$$

Second,

2311.14257_separable_essential_ideal_ubcp_reduction

proved that every C^* -algebra containing a separable essential ideal has UBCP. The goal of this packet is to test how close BCP itself comes to producing such an ideal.

Faithful State Consequence

Theorem 1 (BCP gives a faithful state). *Let A be a C^* -algebra. If A has BCP, then A admits a faithful state.*

Proof. The zero algebra is trivial, so assume $A \neq 0$. Let

$$S_A \subseteq \bigcup_{n=1}^{\infty} B(c_n, r_n)$$

be a BCP cover by open balls, with $r_n < \|c_n\|$. Ignore any redundant zero centres. For each n , choose a state φ_n on the unitization of A such that

$$\varphi_n(c_n^* c_n) = \|c_n\|^2.$$

This is possible because states norm positive elements.

Define bounded linear functionals on A by

$$f_n(x) = \varphi_n(c_n^* x).$$

We claim that the family (f_n) separates points of A . If not, choose nonzero $x \in A$ with $f_n(x) = 0$ for every n , and put $u = x/\|x\|$. Let

$$(\pi_n, H_n, \xi_n)$$

be the GNS representation of φ_n . Then

$$\langle \pi_n(u) \xi_n, \pi_n(c_n) \xi_n \rangle = \varphi_n(c_n^* u) = 0.$$

Hence

$$\begin{aligned} \|u - c_n\|^2 &\geq \|\pi_n(u - c_n) \xi_n\|^2 \\ &= \|\pi_n(u) \xi_n\|^2 + \|\pi_n(c_n) \xi_n\|^2 \geq \|c_n\|^2. \end{aligned}$$

Therefore $\|u - c_n\| \geq \|c_n\| > r_n$ for every n , contradicting that the balls cover S_A .

Thus the f_n 's separate points. Put

$$\psi = \sum_{n=1}^{\infty} 2^{-n} \varphi_n|_A.$$

After normalizing if necessary in the nonunital case, ψ is a state on A . We show it is faithful. If $a \in A$ and $\psi(a^* a) = 0$, then

$$\varphi_n(a^* a) = 0 \quad (n \geq 1).$$

In the GNS representation, this means $\pi_n(a) \xi_n = 0$. Hence

$$f_n(a) = \langle \pi_n(a) \xi_n, \pi_n(c_n) \xi_n \rangle = 0 \quad (n \geq 1).$$

Since the f_n 's separate points, $a = 0$. Thus ψ is faithful. □

Corollary 1. *Every C^* -algebra without a faithful state fails BCP.*

Essential Ideal Generated by a Blocker

Theorem 2 (BCP blockers generate essential ideals). *Let A be a nonzero C^* -algebra with BCP, and let $C \subset A$ be a countable set such that for every $u \in S_A$ there is $c \in C$ with*

$$\text{dist}(c, \mathbb{C}u) < \|c\|.$$

Let $I \triangleleft A$ be the closed two-sided ideal generated by C . Then I is essential.

Proof. Suppose I is not essential. Then there is a nonzero closed two-sided ideal $J \triangleleft A$ with $I \cap J = \{0\}$. For ideals in a C^* -algebra, this implies

$$IJ = JI = \{0\}.$$

Choose $u \in J$ with $\|u\| = 1$.

For every $c \in C$ and every scalar λ , the element $c - \lambda u$ lies in the orthogonal C^* -direct sum $I \oplus J$. Therefore

$$\|c - \lambda u\| = \max\{\|c\|, |\lambda| \|u\|\} \geq \|c\|.$$

Taking the infimum over $\lambda \in \mathbb{C}$ gives

$$\text{dist}(c, \mathbb{C}u) = \|c\| \quad (c \in C).$$

This contradicts the assumed blocker property of C . Hence I is essential. □

Relation to the UBCP Bridge

Combining Theorem 2 with the exact BCP/countable-capture equivalence gives:

Corollary 2. *Every C^* -algebra with BCP has a countable subset whose generated ideal is essential.*

This is close to the previous UBCP theorem, but it is not the same as having a separable essential ideal. The distinction is real:

Remark 1 (why countable generation is not enough). *Let H be a nonseparable Hilbert space and let p be a rank-one projection in $\mathcal{B}(H)$. The closed ideal of $\mathcal{B}(H)$ generated by p is $\mathcal{K}(H)$, which is essential in $\mathcal{B}(H)$ but nonseparable. Thus an ideal generated by one element inside a larger C^* -algebra need not be separable.*

This example does not give BCP: $\mathcal{B}(H)$ fails BCP by the von Neumann algebra classification in the source paper. It only shows that the last missing bridge cannot be reduced to the formal phrase “countably generated ideal.”

What Remains

The packet

`2311.14257_separable_essential_ideal_ubcp_reduction`

proves:

$$\text{separable essential ideal} \implies \text{UBCP}.$$

The present packet proves:

$$\text{BCP} \implies \text{a countable blocker generates an essential ideal}.$$

The still-open bridge is whether the special geometry of a BCP blocker forces enough separable local structure inside that essential ideal to construct a uniform countable cover. Equivalently, a hypothetical BCP-not-UBCP C^* -algebra would have a faithful state and an essential ideal generated by a countable BCP blocker, but no separable essential ideal of the kind used in the previous positive theorem.

Novelty and Search Bounds

Local run indexes were searched for arXiv:2311.14257, BCP, UBCP, faithful state, weak-star separability, essential ideal, countable blocker, and separable essential ideal. Web searches on June 17, 2026 for “ball-covering property faithful state C^* -algebra” and variants did not locate this exact C^* -specific consequence. The density-continuum packet already recorded a Banach-space countable-separating-functional consequence of BCP; Theorem 1 upgrades that mechanism to a faithful state in the C^* -algebra setting.

Human Review Focus

Reviewers should check:

1. the GNS Pythagorean estimate in Theorem 1;
2. that the resulting functionals really separate all points of A ;
3. the faithfulness of the convex combination of the states φ_n ;
4. the orthogonal direct-sum norm calculation in Theorem 2;
5. the limitation in Remark 1.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.
- [3] Bruce Blackadar, *Operator Algebras: Theory of C^* -Algebras and Von Neumann Algebras*, Encyclopaedia of Mathematical Sciences 122, Springer, 2006.